

Revisiting diachronic change in the nominal domain from Latin to modern Romance*

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Abstract

The intuition behind this offering is classical and familiar (cf. Schwegerler 1990): morphological richness of nominal inflection underlies the possibility of Left Branch Extraction in Latin (traditionally called hyperbaton) while the loss of morphological richness of nominal inflection yielded the Left Branch Condition that characterizes modern Romance. Specifically, analogous to the notion “strong” and “weak T” in Chomsky 2015; I here adopt the Nominal Strength Parameter (Blümel 2024), a new instantiation of the classical Borer-Chomsky conjecture which localizes syntactic variation in properties of functional heads. It states

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that the functional nominalizing head n (cf. Borer 2005) comes in two kinds for the purposes of identification of a category label – strong and weak. These two lexical values morphologically correlate with rich (gender, number and case) noun inflection on the one hand, and poor noun inflection on the other. Crucially, the analysis unifies the mentioned syntactic properties pertaining to the distribution of determiner categories with their optionality in Latin and their obligatoriness in modern Romance (cf. e.g. Longobardi 1994 on Italian). This paper makes the novel proposal that the relevant diachronic change in the nominal domain involved a resetting of the value of n from strong to weak.

Keywords: Left Branch Extraction, Left Branch Condition, determiners, articles, noun phrase, labeling, weak vs. strong

1 Introduction

In this paper I reinterpret a classical impression pertaining to the diachrony of Latin towards modern Romance (MR), confining myself here to French, Italian, and Spanish. Generally, this impression ties the loss of morphological richness on various elements to a gradual increase in syntactic rigidity, i.e. a loss of word order freedom in the transition from Latin to MR. Roughly, the transition from what we might descriptively call a non-configurational syntactic profile in Latin gave way to the configurational syntactic systems that characterizes MR. Much recent work has shown that Latin underlyingly is configurational (cf. Devine and Stephens 2006; Ledgeway 2012; Danckaert 2017), i.e. the term non-configurational is taken to be descriptive so as to

classify the surface manifestation that distinguishes Latin from MR.

More specifically and focussing on the nominal domain, the intuition draws a connection between the diachronic morphological impoverishment of nouns on the one hand (cf. Pinkster 1990 on the loss of case in relation to the diachrony of prepositions), and two twin syntactic properties of Latin and MR: Latin did not feature articles nor obligatory determiners and allowed for Left Branch Extraction (LBE); with the diachronic loss of case inflection, an article system developed in MR, going hand in hand with the Left Branch Condition (LBC, cf. Ross 1967). While this is old news, the analysis in which I will couch the described intuition is not. I will implement it by claiming that the nominalizing functional head n (cf. Borer 2005) is parametrized as strong and weak in the sense of Labeling Theory (cf. Chomsky 2013, 2015), building on Blümel (2024) who applies this analysis synchronically. Latin featured the former value while the languages of MR feature the latter value. The distinctive syntactic characteristics, i.e., the distribution of determiner (DET) categories which distinguish Latin and MR follow from this parametrization. Likewise, the lexical parameter morphologically manifests itself in the different degrees of morphological richness of nouns in Latin and MR, specifically in the degree to which gender, number and case forms in the nominal paradigm vary or fail to vary (syncretism). In this sense, this paper supports a traditional view and makes a theoretical contribution, which captures some of the phenomena most prominently captured under the NP-DP-divide (cf. Bošković 2005a *et seq*).

This paper is structured as follows. Section 2 lays out the basic empirical facts to be accounted for. Section 3 presents the theoretical framework and

assumptions for the analysis – which is proposed in Section 4. In Section 4 I also remark on some conceptual ramifications. Finally, the paper is concluded in Section 5.

2 Setting the empirical stage

The empirical facts to be captured pertain to the DET-system in Latin and MR. The pattern is simple enough. Needless to say that with this coarse of an empirical picture in mind, much variation must be glossed over.¹ By DET I mean to include demonstratives, WH-determiners, quantifiers, numerals, possessor and the like. As before, I abstract away from important differences between these categories insofar as their differences do not impinge on the analysis.

As is well-known, Latin allowed for the extraction of a DET-category as illustrated by way of extraction of a demonstrative in (1):

- (1) **hanc_i** cum habeat [**t_i praecipuam laudem**].
 this.ACC since has particular.ACC merit.ACC
 ‘since he has this particular merit’ (Cicero, Brutus 261)

In the literature, this type of extraction is known as LBE. Previous studies of the phenomenon in Latin include Devine and Stephens 2006; Spevak 2010 and Powell 2010 which to some extent highlight information structural and pragmatic effects, which the phenomenon accompanies. Analytically, this

¹For example, I will not address questions like why kind terms in one Romance language requires the presence of an overt determiner while another one does not (while possibly requiring a silent one). For establishing the basic theoretical point, the abstraction is required and, I think, appropriate. I hope that by establishing a basic theoretical approach the nominal structures in Romance, such questions can be seen from a fresh light.

transition has been discussed in terms of the NP-DP-parameter (Bošković 2005a,b, 2008), of “a change in the head-directionality parameter with a concomitant shift from specifier- to head-oriented syntax in the domain of functional structure on the other” cast in terms of roll-up derivations (Ledge-way 2012), and of phonological movement in conjunction with several other phenomena (Agbayani and Golston 2016). These analyses all have great merits. However, one aspect which in my view is crucial must be captured by any account: The connection between LBE and morphologically rich concord on the noun and DET to license it – an aspect which some works acknowledge, but few capture as far as the analysis goes.

Latin’s LBE option contrasts with the absence thereof in MR, an array of languages which are subject to the LBC, a term which appeals to the linear position of the category Ross (1967); later updated to include the category immediately dominated by NP by Chomsky (1973, fn. 10); a view which subsequently received a lot of independent and cross-linguistic support, cf. e.g. Grosu (1974). The LBC is anachronistically formulated in (3)² and exemplified by AP-extraction in Italian in (2-a). Of course, the occurrence of AP contiguous with the noun is well-formed, cf. (2-b):

- (2) a. *chi [più alta] fra di loro ha [la t_i autorità].
 who more high among of them has the authority
- b. chi fra di loro ha [la più alta autorità].
 who among of them has the more high authority
 ‘who has the highest authority among them’. (Italian;

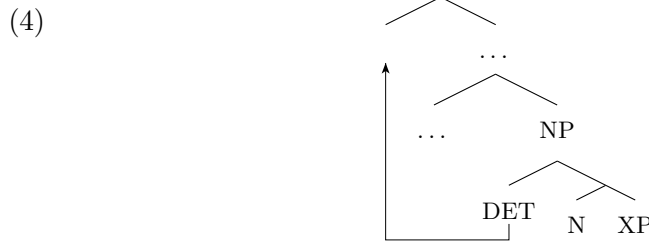
²I retain this formulation here, because it resembles more closely the analysis proposed in this paper, compared to formulations committed to the DP-hypothesis. For recent discussion of general issues, cf. Blümel and Holler 2022. For specific issues of that empirical domain related to labeling, cf. *op. cit.*; fn. 9.

Ledgeway 2012, 424)

(3) **The Left Branch Condition (LBC);** Ross (1986, 127)

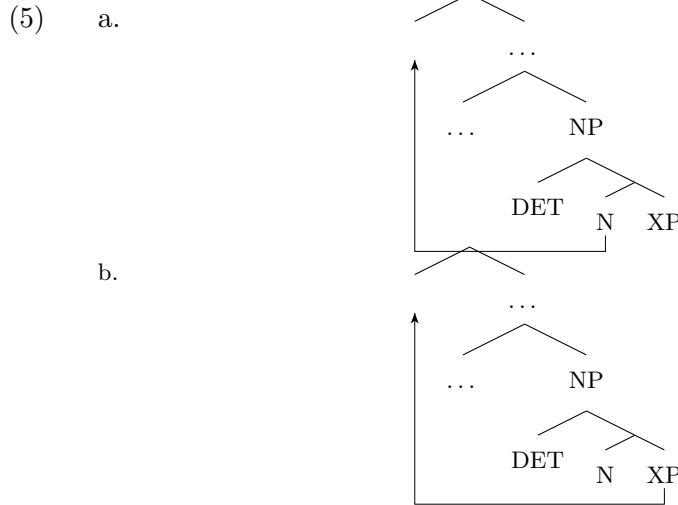
No NP which is the leftmost constituent of a larger NP can be re-ordered out of this NP by a transformational rule.

To avoid misunderstandings and by way of the tree graph (4); LBE involves extraction of DET in the languages that allow it, and the LBC bans precisely that type of movement in the languages that disallow it:



LBE does not designate movement of N as shown in (5-a) or subextraction of XP, the complement of N, as shown in (5-b). At least (5-b) is independently attested in Latin but is plausibly dissociated from the possibility of LBE (e.g. English allows (5-b) but disallows LBE). As far as (5-a) is concerned, whether or not Latin features this type of movement remains to be shown.³ I will briefly get back to extraction of postnominal genitives in Latin in Section 4.2; i.e. an instance of (5-b); but will be primarily be concerned with LBE in this paper.

³I refer the reader to Ott (2011) for compelling arguments that German ‘split topics’ do not involve (5-a); impressions notwithstanding.



The island constraint LBC is a deep-seated puzzle, and various attempts to derive it exist in the literature. That languages like Latin disobey the constraint was noticed by Ross (1967) at once. Many researchers have pointed out agreement as one crucial factor involved in the possibility of LBE, i.e., of circumventing the LBC (cf. Ross 1967; Corver 1990, 2010; Zlatić 1997, ch. 5; Bošković 2013a; Reeve 2019). For instance, being concerned with Ancient Greek and using a schematic hyperbaton structure Y_1XY_2 , Devine and Stephens (2000, 224) characterize the relation as follows: “the morphology serves as an overt coindexing system that guides unification of the discontinuous Y elements into a single semantic constituent.” I will elaborate on this aspect in the course of the discussion.

Another well-known observation is that MR encompasses languages which require the presence of a DET-category, i.e. DET is obligatory. While the empirical observation is restricted to singular count nouns, mass nouns and bare plurals are analytically treated along the same lines, i.e., with an oblig-

atory phrase DET which remains silent. The seeming optionality of DET in bare plurals and mass nouns is thus traced to a null-variant of DET in these contexts. In this sense, I follow conventional treatments within the DP-hypothesis, albeit with the qualification that the mass/plural operator is phrasal. The obligatory nature of DET in MR is illustrated by (7)-(9); though it should be noted that French even disallows bare plurals.⁴

- (7) *(Un/il) Grande amico di Maria mi ha telefonato.
 (a/the) great friend of Maria me have called-up
 ‘(A/the) good friend of Mary called me up.’ (*Italian*; Longobardi 1994, 612)
- (8) Limpio #(el) coche.
 clean (the) car
 ‘I’m cleaning (the) car.’ (*Spanish*; Espinal 2010, 993)
- (9) a. *Etudiants ont occupé l’édifice.
 students have occupied the building
 b. *J’ai vu étudiants dans l’édifice.
 I have seen students in the building
 (*French*; Delfitto and Schroten 1991, 156)

⁴The facts must be treated with much care. For example, bare nouns which denote properties of kinds are possible in Spanish as (6) shows.

- (6) a. Lleva jersey.
 wears pullover
 ‘(S)he is wearing a pullover.’ (It could be one, or more than one)
 b. Comprará coche.
 buy-FUT car
 ‘(S)he will buy a car.’ (It could be one, or more than one) Espinal (2010, 985)

I abstract from such –and more– complications, like the difference between pre- vs. postverbal positioning of arguments in the Romance pro-drop languages. That there exists an elementary difference between the nominal systems of MR and the one of Latin is not disputed.

Once more, Latin contrasts with this observation in that DET is optional. This is illustrated by the lack of anaphorically used nouns in (10); where *legatos* ‘emissaries’ introduces the referent and subsequent mentions of that referent must hence be anaphorically definite. No DET-category is required, unlike in the English translation and, importantly, unlike counterparts in MR.

- (10) *legatos ad eum mittunt ... legatis respondit. ... ubi ea dies quam constituerat cum legatis venit et legati ad eum reverterunt ...*
 ‘They sent emissaries to him ... He answered the emissaries ... When the day came that he had set up with the emissaries, and the emissaries came back to him ...’ (Caes. B.G. 1.7-8 in Wigtil 1993, 468)

In this connection, it is worth mentioning that Latin lacked articles altogether, a property often associated –but not identical to– with observations like (10). Thus Ledgeway (2012, 424) notes that

“Quintilian’s (Inst. 1.4.19) oft-cited observation ‘our language does not need articles’ might be taken to mean that not only did Latin lack articles, but that it also lacked a dedicated position for D(eterminers).”

When analyzing the observed pattern I will return to an interpretation of the relation between optional DET and the absence of articles in Latin. As before, the link between the availability of articles and the LBC being operative is well-established (cf. cf. Uriagereka 1988, 115; Corver 1990,

2010; Bošković 2005a, 2012 *et seq*⁵). The resulting one-way correlation at a descriptive level runs as follows:

- (11) a. If a language has articles, it is subject to the LBC.
b. If a language lacks articles and lacks nominal concord, it is subject to the LBC.
c. The only way for a language to be exempt from the LBC is if it lacks articles and exhibits nominal concord.

Now, I take this correlation to be well established. But an important question and lacunae is if the absence of articles and the presence of nominal concord are sufficient condition for a language to have LBE. The answer is clearly no. Estonian is a straightforward counterexample since it lacks articles (cf. Norris 2014, 33), exhibits nominal concord with respect to case, but is subject to the LBC, cf. (12)-(13):

- (12) a. Heiko müü-s uue auto maha.
H.NOM sell-PST.3.SG new.ACC car.ACC PRTCL
'Heiko sold a/his new car.'
- b. [Uue auto]_i müü-s Heiko *t_i* maha.
new.ACC car.ACC sell-PST.3.SG H.NOM PRTCL
'The new car, Heiko sold.'
- c. *Uue_i müü-s Heiko [*t_i* auto] maha.
new.ACC sell-PST.3.SG H.NOM car.ACC PRTCL
'The new car, Heiko sold.'
- Norris (2014, 30)

⁵But see Pankau (2019) for the potential counterexample Lower Sorbian, which however, is a language in close contact with the dominant language German – possibly a confounding factor.

- (13) a. Sa luge-si-d Kärdi raamatu-t.
 you read-PST-2.SG Kärt.GEN book-PAR
 ‘You read Kärt’s book.’
- b. *Kelle_i sa luge-si-d [_i raamatu-t]?
 who.GEN you read-PST-2.SG book-PAR
 Intended: ‘Whose book did you read?’
- c. [Kelle raamatu-t]_i sa luge-si-d _i?
 who.GEN book-PAR you read-PST-2.SG
 ‘Whose book did you read?’ (*Estonian*; Norris 2014, 30-31)

Furthermore, the term nominal concord is used in different ways in the literature sometimes encompassing languages which have agreeing modifiers to nouns, where, however, the noun itself does not inflect for the relevant features. German is such a language. (14) illustrates different gender forms on adjectives for the strong adjective declension; (15) illustrates different case forms for the strong adjective declension. As can be seen, the noun does not covary morphologically, but remains invariant. While a complete picture is more involved, the picture presented here is adequate enough.

(14) *Gender (Nominative case)*

- a. ein großer Berg
 ein big-M.SG.NOM mountain
- b. eine große Schale
 ein big-F.SG.NOM bowl
- c. ein großes Feuer
 ein big-N.SG.NOM fire

(*German*)

(15) a. *Accusative*

- (i) einen großen Berg
ein big-M.SG.ACC mountain
- (ii) eine große Schale
ein big-F.SG.ACC bowl
- (iii) ein großes Feuer
ein big-N.SG.ACC fire
- b. *Genitive*
 - (i) eines großen Bergs
ein big-M.SG.GEN mountain
 - (ii) einer großen Schale
ein big-F.SG.GEN bowl
 - (iii) eines großen Feuers
ein big-N.SG.GEN fire

(*German*)

I use nominal concord here in a narrower sense in that next to covariation of certain features on modifiers and inherent ones on the noun, the noun crucially needs to exhibit morphological inflection of the relevant feature. This means that German does not exhibit nominal concord in the intended sense, because the nouns for the most part fail to morphologically manifest underlying gender and case differences.

I here amend (11-c) by (16) saying that next to the absence of articles, it is the presence of morphologically realized nominal concord on both the noun and DET with respect to the features number, gender and case which is required for the exemption from the LBC:

- (16) The only way for a language to be exempt from the LBC is if
 - a. it lacks articles and

- b. exhibits nominal concord (on N and DET) with respect to number, gender and case.

By naming all the relevant features which need to be involved in nominal concord a more restrictive morphological precondition for LBE is ensured – as is the case empirically and cross-linguistically. Notice that the “if” in (16) lacks an ‘f’ – and deliberately so: It is not a biconditional but weaker. It states that a necessary condition for LBE is the absence of articles and the availability of nominal concord on N and DET which must include all of number, gender and case. Since it is unidirectional, there might still be languages which have all these properties and still be subject to the LBC. Since Estonian is a language which lacks grammatical gender altogether, it does not fall under (16) and so it is expected that the language disallows LBE.⁶

⁶A poetic style of literary Finnish is mentioned in Bošković 2012 (based on Franks 2007), a language which permits LBE, cf. (17-a). Contemporary colloquial Finnish disallows LBE and has been developing the definite article *se*; cf. (17-b) (Laury 1997):

- (17) a. Punasien_i ostin [_i tauton].
red-ACC buy-PST-1SG car-ACC
(literary Finnish, poetic style)
- b. ?*Punasien_i ostin [(sen) _i tauton].
red-ACC buy-PST-1SG the car-ACC
(spoken Finnish)

The evolution of an article in the language and the seemingly concomitant development of the LBC in the language straightforwardly dovetails with (16-a). Notice a complication, however: Bošković (2016) maintains that what counts as definite article is important. Having the meaning of a iota operator is one criterion, that it has a form distinct from a demonstrative is another one (cf. also Oda 2022, 2-3). However, so conceived the relevance of the colloquial Finnish example is unclear because the article *se* is homophonous with the demonstrative.

This could mean that colloquial Finnish is subject to the LBC due to an independent factor. The absence of gender in Finnish could be one such factor, conforming with (16-b). However, that raises the question why LBE is possible in literary Finnish. The matter

Returning to Latin, a range of categories can undergo LBE in the language. LBE is attested, next to demonstratives as in (1); repeated here as (18-a); for possessors (18-b); WH-phrases (18-c) (Devine and Stephens 2006, 542ff) and APs (18-d). This is reminiscent of the state of affairs in much of Slavic (cf. Bošković 2005a).

- (18) a. hanc_i cum habeat [_{t_i} praecipuam laudem].
 this.ACC since has particular.ACC merit.ACC
 ‘since he has this particular merit’ (Cicero, Brutus 261)
- b. meo_i tu epistulam dedisti [_{t_i} servo].
 my.DAT you.NOM letter.ACC gave servant.DAT
 ‘you gave the letter to my servant?’ (Plautus, Pseudolus 1203)
- c. quis_i umquam [_{t_i} Graecus] comoediam scripsit
 which.NOM ever Greek.NOM comedy.ACC write
 ‘Which Greek ever wrote a comedy?’ (Cicero, pro Flacco 27.65)
- d. qui **summam**_i inter eos habet [_{t_i}
 who highest.ACC.F.SG among them.ACC has
auctoritatem].
 power.ACC.F.SG
 ‘who has the highest authority among them’ (Caes. B.G. 6.13.8; in Bauer 2009, 290)

The paradigm in (19)-(21) gives illustrations of the LBC in MR.

(19) *French*

- a. *Quels_i a-t-il vendu [_{t_i} livres]?
 which has-he sold books
 ‘Which books did he sell?’ (in Corver 2010, 6)

requires further examination.

- b. *Cette_i j'aime [_{t_i} fille]!
 that I love girl
 'That girl I love!' (in Corver 1990, 19)
- c. *Son_i j'ai lu [_{t_i} livre]!
 his I have read book
 'His book I have read!' (in Corver 1990, 19)
- (20) *Questi_i hai incontrato [_{t_i} ragazzi]?
 which you-have met boys
 'Which boys did you meet?' (*Italian*; Rizzi 1982)
- (21) *¿Cuántos_i compró María [_{t_i} libros]?
 how many bought Mary books
 'How many did Mary buy books?' (*Spanish*; Francom 2012, 540)

So an important task is accounting for the change that led from Latin to MR in the nominal domain.

Bošković (2005a) takes the possibility of AP-extraction to be definitional of LBE, and I follow him in this regard. There seems to be an implicational relation between the possibility of AP-LBE and the other DET-categories. That is, I have not found a language which allows AP-LBE but not also all the other DET-categories (wh-determiners, demonstratives, possessors, etc.). The reverse is not true in that languages which allow selected DET-items to undergo extraction never at the same time allow AP-LBE. Thus, Hungarian famously permits extraction of dative possessors (Szabolcsi 1983) but disallows APs to do the same. Much of Germanic allow the extraction of the counterpart *what* in *what for a book* ('what kind of book', cf. Leu 2008, 2014 and references therein) but disallows APs to do the same. French *combien* ('how many') can split off from the associated noun phrase, but

APs cannot (cf. Starke 2001). And so on.

While I subscribe to the view that AP-extraction define LBE-languages (while languages of the exceptional type I just mentioned are not defined this way), a caveat is required: I here abstract away from the analytical treatment of APs, but will here be concerned with the analysis of the other DET-categories illustrated. The reason is that even though AP-extraction makes the right demarcation at a descriptive level, their analysis poses two non-trivial problems with rich conceptual and empirical ramifications: Their mode of being introduced into structure (pair-Merge or set-Merge, cf. Oseki 2014 for an important take at unifying the two) on the one hand and nominal concord for attributive adjectives on the other (for one approach, cf. Norris 2014). The analytical treatment of APs exceeds the limits of this paper, but I hope to address it in the future.

Summarizing, we have seen that DET occurs optionally in complex nominals in Latin and obligatorily so in MR. Moreover, Latin allows LBE of DET-categories but MR does not. Descriptively, we can conclude that external and internal Merge of DET is optional in Latin, while external Merge is obligatory and internal Merge of DET is banned in MR. Before presenting the analysis of this pattern, I will lay out the theoretical background in which it is developed.

3 Theoretical background: Labeling Theory

Recent syntactic theorizing is characterized by an effort to reduce the ingredients of the operation which builds hierarchical structure called Merge (cf.

Chomsky 1995). These efforts have arguably reached their peak, reflected in the superlative of the term Simplest Merge, cf. Epstein et al. (2014). A parallel development has moved towards progressively liberating the application of Merge from the dominant role that features play in syntactic derivation (Chomsky 2004, 2013, 2015; the latter two works will be referred to as POP/+). A fundamental assumption that has emerged from these tendencies of inquiry is that the application of Merge is not feature-driven – while the representations emerging from its application are to an extent subject to conditions which lexical properties determine. Specifically, (both external and internal) Merge applies freely, untriggered, as shown schematically in (22); which is also meant to illustrate (a) that no linear ordering is imposed on the elements α and β and (b) that no projection mechanism is built into the operation.

$$(22) \quad \text{Merge}(\alpha, \beta) \rightarrow \{\alpha, \beta\}$$

Of course, for this view to be tenable descriptively, there must be constraining factors to prevent overgeneration. By assumption, these factors are extraneous to narrow syntactic computation. Possibly and as suggested in POP/+, these factors apply only at the point when the cyclic operation Transfer hands over syntactic representations to the interfaces in a phase-by-phase fashion. The specific operationalizations that POP/+ proposes are given in (23-a) and (23-b):

- (23) a. the requirement that syntactic objects be labeled, informing the question: What kind of object is it?

- b. the 3rd factor principle of computational efficiency *Minimal Search*; which detects labels in an efficient manner, i.e., structurally closer is better than remote (POP: 43).

I will henceforth refer to (23-a) as the Labeling Requirement LR. To reiterate: Merge is maximally simple, i.e., just (22). It symmetrically combines two elements within an unordered set and without any encoding of projection and thus no notion of specifier.

To meet the LR, POP proposes a Labeling Algorithm (LA) which finds the most prominent element in a given set. The guiding intuition behind the LA is that structurally closer wins out over structurally remote. Whenever a simple element Merges with a complex one, the simple one wins out. When the two Mergees are equally complex, more needs to be said. The three logically conceivable cases below illustrate the LA with specific examples. Underlining is used to indicate the element found as a label:

Head-XP: LA detects the head as the label (XP requires deeper search).

$$(24) \quad \{V, NP\} = \{V, \{N, \dots\}\} \rightarrow \{\underline{V}; NP\} = VP$$

Head-Head: The label is the categorizer which bears grammatical features, not the root R, which does not bear grammatical features and is thus “too weak” to be found as a label.

$$(25) \quad \{v; R\} \rightarrow \{\underline{v}; R\} = vP$$

XP-YP: The label is either

- YP iff XP undergoes internal Merge (i.e., moves)

E.g., intermediate configurations in successive-cyclic movement of the format

$$(26) \quad \{\alpha \text{ XP, CP}\} \rightarrow \text{XP} \dots \{\text{CP} \langle \text{XP} \rangle, \{\underline{\text{C}} \dots\}\}$$

- or $\langle f, f \rangle$, where X and Y bear f and agree with respect to f .

(27) the target of EPP-raising in English:

$$\{\text{NP, TP}\} \rightarrow \{\langle \varphi, \varphi \rangle \{\text{N}_{\underline{\varphi}} \dots\}; \{\text{T}_{\underline{\varphi}} \dots\}\}$$

The option (27) implicates an innovation compared to classical $\bar{\text{X}}$ -theoretic and Bare Phrase Structure-conceptions of projecting a category tag: The LA can target features, not only Lexical Items.⁷

The derivation of a simple finite sentence comprising a transitive verb phrase in (28) illustrates the working of the LA, including EPP-raising of the subject-NP. For example, *Susan kicks the ball*; where NP=*Susan*; $v\text{P}$ =*kick the ball* and T=[Tense: PRS][$u\varphi$:--]:

- (28) a. $\{? \text{NP, } v\text{P}\}$ labeling problem; (external) Merge T
 b. $\{\text{T, } \{? \text{NP, } v\text{P}\}\}$ (internal) Merge NP
 c. $\{\text{NP, } \{\text{T, } \{v\text{P} \langle \text{NP} \rangle, \{v; \dots\}\}\}\}$ labeling possible
 d. $\{\langle \varphi, \varphi \rangle \text{ NP, } \{\text{T, } \{v\text{P} \langle \text{NP} \rangle, \{v; \dots\}\}\}\}$

⁷Cf. POP: 45: “it must be that LA seeks features, not only LIs – or perhaps seeks only features, in which case it would be similar to probe-goal relations generally, specifically Agree.” The current approach to NCs extends this line of thinking.

In (28-a) there is a problematic symmetry in that neither N nor *v* can provide the label of NP-*v*P. Assuming that NP raises upon Merge of T, NP-*v*P receives a label by *v*; because by assumption, the lower copy of the subject-NP is invisible to LA (for an intriguing deduction of this postulate, cf. Epstein et al. 2020). NP-TP in (28-c) is labeled by the common feature borne by N and T, namely φ . The result is shown in (28-d). This gives the desired results whenever a transitive structure is involved.

What of a *general* deduction of the EPP-property in English, including the obligatory raising of grammatical subjects in, e.g., passives and unaccusatives? To solve this problem, POP+ introduces the notion of “weak T” and “strong T.” The intuition behind this distinction is that there the morphological richness of verb inflection signals an underlying lexical parameter of T with significant consequences for the syntactic computation and the LA. Thus weak T underlies the poorly inflecting verbal φ -features in English. Weak T is in effect a defective category to label α in (29-a). Raising of the internal argument needs to apply to “support” weak T so that a $\langle\varphi, \varphi\rangle$ -label can be identified by the LA as in (29-b):

- (29) a. $*[\alpha T_{wk}=was [kicked\ the\ ball]]$
 b. $[\langle\varphi, \varphi\rangle [NP\ the\ ball]_i T_{wk}=was [kicked\ t_i]]$

As before the feature shared between N and T results. This time, however, EPP-raising obtains not for a problematic symmetry in the base structure but because of the defective nature of weak T to provide the label for its containing set.

POP+ shows that this conception makes a unification of the EPP and the Empty Category Principle (ECP) possible. Consider the *that*-trace effect in (30). From the perspective of Labeling Theory, $\bar{A}$ -moving a subordinate subject-NP restores the problematic unlabelability of α in (30) due to the invisibility of the “trace” for the LA:

- (30) **Who_i does Benjamin think that [_αt_i T_{wk} kicks the ball]*

Unlike English, null-subject languages like Italian with rich verbal agreement paradigm have “strong T.” As a consequence, there is no need (a) to raise an NP to the sister position of TP, and (b) to retain the subject within the subordinate clause. The uniform reason is that strong T can label the structure without support. This, according to POP+, is the underlying cause for pro-drop (31-a) and for the absence of ECP-effects (31-b) in Italian and comparable pro-drop languages:

- [illegible]

The upshot of this exposition is this: Based on the freely applying operation Merge and in interaction with 3^{rd} factor principles and interface requirements, diverging syntactic effects derive from a single parametric property on the functional head T: its strength and weakness.⁸

⁸There are numerous issues surrounding this proposal, cf. Goto 2017; Kruger 2017;

This should suffice as theoretical background and we are now in a position to develop the current analysis. In a nutshell, I propose that the conception of parametrizing T for the purposes of labeling can be fruitfully carried over to the nominal domain.

Hayashi 2020; Obata 2018 and Yanagisawa 2021 and several works cited therein dealing with empirical and analytical aspects surrounding the matter. A particular empirical issue concerns the analytical treatment of the familiar contrast between the ungrammatical (30) and the grammatical (32) which involves a null-complementizer.

- (32) *Who_i does Benjamin think [_αt_i T_{wk} kicks the ball]*

The exposition here cannot do justice to the complexity of the matter. A particularly interesting approach is provided by McInnerney (2024) who gives two modifications of the POP+ systems. POP+ assumes (33-a) and (34-a), while McInnerney (2024) proposes the alternatives (33-b) and (34-b):

- (33) The presence of an unlabeled syntactic object x at the interfaces leads to a crash
 ...
 a. immediately.
 b. at the next application of Transfer.
- (34) When labeling applies at the phase level, Search targets ...
 a. the full phase.
 b. the transferred domain.

He gives empirical arguments showing that in English, *wh*-subjects must pass through SPEC-CP belonging to the clause where they originate in questions like (30). This is contrary to the assumption in POP+ where crucially, movement from SPEC-TP to SPEC-CP is disallowed (cf. also Chomsky 2008; Obata and Epstein 2011). Analytically McInnerney (2024) appears to implicate (cf. *op cit.*, 592), in effect, that *that*-trace effects are phonologically ill-formed but syntactically well-formed – which strikes me as reasonable, given not just the relevant empirical evidence from English. He is concerned with the syntactic aspects, though, and develops an analytical alternative to the approach to subject *wh*-questions in POP+, based on the modifications (33-b) and (34-b) leaving the other assumptions of the labeling system intact.

An analogous treatment in the nominal domain would entail that the LBC is phonological in nature, not syntactic. An exciting prospect, whose execution, however, must be left for future work.

4 Proposal: nominal strength and weakness

In this section I suggest that analogous to the strong/weak distinction on T, the nominalizing functional head n should be parametrized. That is, we expect to find nominal strength and weakness. Let us briefly go back to the previously mentioned common observation that LBE goes hand in hand with rich morphology while the LBC does not: “As was already hinted at by Ross [1967], languages violating the LBC are characterized by rich morphology” (Corver 2010). The observation raises the question which element it is which exhibits “rich morphology.” The central hypothesis in Blümel 2024 is given in (35); and I apply it here to the languages listed:

(35) **The Nominal Strength Parameter (NSP)**

- a. strong n/n_{str} : Latin
- b. weak n/n_{wk} : French, Italian, Spanish . . .

Implied is the conception of functional heads which categorize category-neutral roots (cf. Marantz 1997; Borer 2005), here n .

What is the empirical motivation for parametrizing n ? The classical intuition going back to at least Schwegler (1990) is that the morphological richness of Latin nominal inflection is tied to its LBE characteristic. More specifically, the gradual change in nominal morphology concerned gender and case inflection. Classical Latin (the Old Latin period until around 75 BC) had the three values masculine, feminine and neuter within its gender system and lost neuter in later stages, often associated with the term Vulgar Latin, having both historical and sociological aspects. The remaining femi-

nine/masculine gender system on nouns permeated MR.⁹ Likewise, the rich case system of Latin nouns exhibited –depending on the exact counting– 5-8 values with at least nominative, accusative, dative, genitive and ablative, and possibly vocative, locative and instrumental in addition. These were lost on nouns entirely in MR.

The loss of morphological richness of nominal inflection yielded the LBC that characterizes MR. Abstracting away from declension classes, let us rehash the nominal inflection and case paradigms in Latin, according to (35-a) a strong *n* language, and compare it the morphological properties of the MR languages which are set to the weak side of the parameter, cf. (35-b). The comparison is straightforward as the table in (36) shows in relation to (37):

(36) **The strong**

Latin; sg.	1.F	2.M	2.N
NOM	<i>-a</i>	<i>-us</i>	<i>-um</i>
GEN	<i>-ae</i>	<i>-i</i>	<i>-i</i>
DAT	<i>-ae</i>	<i>-o</i>	<i>-o</i>
ACC	<i>-am</i>	<i>-um</i>	<i>-um</i>
ABL	<i>-a</i>	<i>-o</i>	<i>-o</i>

(37) **The weak**

- French: No nominal case inflection

⁹But see Loporcaro (2017) for a more nuanced and comprehensive view.

- Italian: No nominal case inflection
- Spanish: No nominal case inflection

Against the backdrop of the observations in the sentential domain in Section 3; we can establish the following descriptive claim, assuming that the obligatory presence of DET-categories is an EPP-effect:

- (38)
- a. The richness of nominal inflection is the nominal counterpart of the richness of verbal inflection. Just as the latter licenses (a) pro-drop and (b) circumvents ECP-effects, the former licenses (a') DET-drop and (b') circumvents ECP-effects.
 - b. Poverty of noun inflection is the nominal counterpart of the poverty of verbal inflection. Just as the latter gives rise to a (a) EPP-effect and (b) ECP-effects in the clause, the former gives rise to (a') EPP-(=obligatory DET-)effects and (b') ECP-effects in the nominal domain.

4.1 Executing the analysis and initial predictions

In line with the head-head condition in (25); a minimal NC is (39):

- (39) $\{n; R\}$

R is a category-neutral root, constituting the open class elements. Depending on the functional head R Merges with, it will be categorized differently:

- (40) a. the big $[n \sqrt{house}]$

- b. They [$v \sqrt{house}$ me].

n is a nominalizing functional head (cf. Marantz 1997; Borer 2005). I will assume that it bears gender, number and Case features.

$$(41) \quad n_{[GEN, NUM, CASE]}$$

An additional assumption of the analysis is that determiners are phrasal (Sag et al. 2003; Chomsky 2007, 25-26; Oishi 2015). For the purposes of this paper and following Leu (2008, 2014); I will assume that DET-categories are adjectival (xAP, extended adjectival phrase), headed by xA.¹⁰ As an effect, Merging [n R] and xAP yields (42); an instance of XP-YP:

$$(42) \quad \{\text{xAP}, \{n; \text{R}\}\}$$

Without any additional assumptions, the NSP makes the following six predictions pertaining to the distribution of DET categories in Latin and MR:

$$(43) \quad n_{str} \text{ (Latin)}$$

- a. [$_{\alpha} n_{str}$ R]
- b. [$_{\alpha}$ xAP [n_{str} R]]
- c. xAP _{i} . . . [$_{\alpha} t_i$ [n_{str} R]]

$$(44) \quad n_{wk} \text{ (MR)}$$

- a. $^*[_{\alpha} n_{wk}$ R]
- b. [$_{\alpha}$ xAP [n_{wk} R]]

¹⁰Generalizing a view that was initially confined to Slavic, cf. Horn (1978, 1983); Zlatić (1997); Bošković (2005a).

c. *xAP_i... [_α t_i [n_{wk} R]]

The LA labels α in (43-a) by n_{str} which empirically translates into the observation that in Latin, determiners are optional. The LA labels α in (43-b) by a feature n_{str} and xA share. This will accommodate an aspect of nominal concord in Latin as I will elaborate momentarily. The LA labels α in (43-c) by n_{str} which captures the familiar observation that LBE is possible in Latin. As can be seen, all structures meet the LR, because n_{str} supports a label throughout. After having considered the predictions as far as the n_{str} language Latin goes, let us consider the n_{wk} languages of MR.

The LA cannot label α in (44-a) because n_{wk} and R are too weak to be detected by the LA. This captures the observation that MR obligatorily requires the presence of a DET category, now interpreted as a violation of the LR. The LA labels α in (44-b) because n_{wk} receives “labeling support” by xAP: They share a prominent feature. This is analogous to what POP+ suggests for the EPP in English-type languages. Finally, the LA has to leave α in (44-c) unlabeled because n_{wk} and R are too weak to label and the trace is invisible to LA. Specifically, taking (20) repeated as (45) for illustration, if *Minimal Search* targets β in (46), it finds xAP1 and will fail to find the identical copy xAP2 (numbering for exposition only), leaving α fatally unlabeled.

(45) * [_β Questi_i hai incontrato [_α t_i ragazzi]]
 which you-have met boys

(46) [_β xAP1 [C ... [_α xAP2 [n_{wk} R]]]

This aspect of n_{wk} derives the LBC in MR from the LR. The next subsection addresses the question which respective labels obtain.

As we have seen, core data from the distribution of DET that distinguished Latin from MR are captured by parametrizing n : n_{str} in Latin yields optional DET and LBE, while n_{wk} in MR yields obligatory DET and the LBC. The diachronic passage from Latin to MR which suggests itself is thus one in which the parameter is turned from strong to weak with the concomitant syntactic effects:

$$(47) \quad n_{str} > n_{wk}$$

This resetting of the parameter of n went hand in hand with a loss of gender and case inflection on nouns – a well-established observation in the philological literature. Of course, the idea is that this loss of case inflection was a morphological reflex of the lexical change embodied in (47).

Now let us refine both predictions and analysis.

4.2 Further predictions and refinements of the analysis

Ledgeway (2012) mentions other important ways in which Latin differed from MR in the nominal domain. He observes that placement of possessive adjectives can be prenominal (48-a) and postnominal (48-b) in Latin. (49) illustrates the variable positioning for lexical possessives.

$$(48) \quad \begin{array}{llll} \text{a.} & \text{Caesar} & [\alpha \text{ suas} & \text{copias}] & \text{subducit} \\ & \text{Caesar}_{NOM} & \text{his}_{ACC.FPL} & \text{troops}_{ACC.FPL} & \text{withdraws} \\ & & & & (\text{Lat., Caes. B.G. 1.22.3}) \end{array}$$

- b. [_{α} copias suas] Caesar [...] subducit
troops_{ACC.FPL} his_{ACC.FPL} Caesar_{NOM} withdraws.
(Lat., Caes. B.G. 1.24.1)
- (49) a. Nam ne eius supplicio [_{α} Diuiciaci animum] offenderet, uerebatur.
“For he was afraid that he should hurt the feelings of Divitiacus by the punishment of this man.” (Lat., Caes. Gall. 1.19.2 in Viti 2010, 83)
- b. Hi consuetudine [_{α} populi Romani] loca capere, castra munire, commea tibus nostros intercludere instituunt.
“These, adopting the practice of the Roman people, begin to select advantageous places, to fortify their camp, to cut off our men from provisions.” (Lat., Caes. Gall. 3.23.6 in Viti 2010, 85)
- (50) a. César retire [_{α} ses troupes].
Caesar withdraws his troops
‘Caesar withdraws his troops’.
- b. *César retire [_{α} troupes ses].
Caesar withdraws troops his
‘Caesar withdraws his troops’. (French)

The current approach captures this in that n_{str} can invariably label α – it thus does not matter if the possessor appears in “SPEC” of n as in (48-a) or what is arguably a complement position of the possessee noun in (48-b). This contrasts with MR, where possessives must show up in “SPEC”- n_{wk} to support n_{wk} for the identification of α ’s label, as exemplified by (50-a) and (50-b). That is, the deviance of (50-b) derives from the structure’s violation

of the LR.

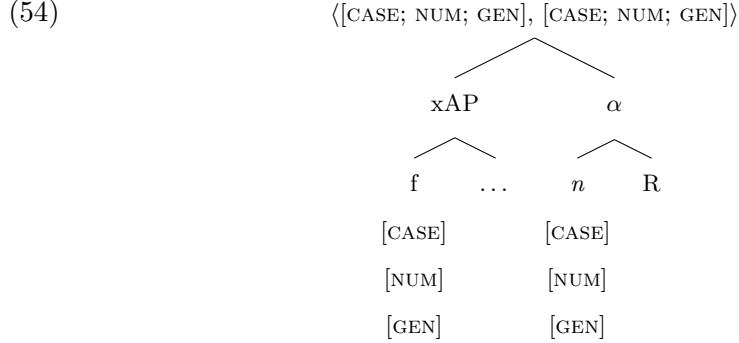
Just what *is* the label of α in (43-b) and (44-b)? Analogous to the approach in Chomsky (2015) for the TP-domain, pairings of features f borne by xA and n suggest themselves as shown in (51-b) and (52-b):

- (51) a. $[\alpha=nP \ n_{str} \ R]$
 b. $[\alpha=\langle f,f \rangle \ xAP \ [n_{str} \ R]]$
 c. $xAP_i \ \dots [\alpha=nP \ t_i \ [n_{str} \ R]]$
- (52) a. $*[\alpha=? \ n_{wk} \ R]$
 b. $[\alpha=\langle f,f \rangle \ xAP \ [n_{wk} \ R]]$
 c. $*xAP_i \ \dots [\alpha=? \ t_i \ [n_{wk} \ R]]$

Latin exhibits nominal concord in case, gender and number as shown by the AP *instabilis* ‘unstable’ and associated noun *fors* ‘fortune’ in (53-a) and by the demonstrative *hic* ‘this, here’ and the associated noun *splendor* ‘brightness’ in (53-b).

- (53) a. **In-stabilis** in istum plur-imum
 NEG-stable.F.SG.NOM in this.M.SG.ACC much-SUPL
 fortuna valuit.
 fortune.F.SG.NOM strong.3.SG.PERF.IND.ACT
 ‘Unstable fortune has been most powerful against this one.’
 (Rhet. Her. 4.44 in Feldcamp 2021)
- b. **Hunc** tu vitae **splendorem**
 this.M.SG.ACC 2.SG.NOM life.F.SG.GEN brightness.M.SG.ACC
 maculis aspergis istis?
 stain.F.PL.ABL sully.2.SG.PRES.IND.ACT this.F.PL.ABL
 ‘You sully this brilliance of a life with those insults?’

This suggests that the relevant label f of the noun group could be either or all of these features, most of which are abstract in n_{wk} -languages. I will assume so such that the relevant labels are as in (54):



Importantly, these features are not stipulated diacritics but reflect the empirical facts (if the evidence for abstract Case is indirect in some of the languages under consideration). What is a label needed for? I presume that labels are required for interpretation at the interfaces: For example, unordered sets generated by Merge need to be linearized when externalized, and for determining the directionality of the head a label could be required. As for semantic interpretation (54) raises the question what role Case and gender play. Are they both uninterpretable? On the meaning of morphological case, Jakobson (1984) makes highly differentiated and subtle distinctions by way of the Russian case system, critically reviewed and developed in Franks 1995. I here remain agnostic on the matter.

I will assume that the identification of the label is a requirement for semantic interpretation, but not that the label itself needs to be interpreted. As far as Case goes, this is reminiscent of the Visibility Condition which states that an element needs to bear Case in order to be visible for θ -

marking (Chomsky 1986). (54) could be considered a reinterpretation of this condition: A nominal set formed by Merge needs a label to be visible for θ -marking. Case (and gender and number) is such a label.

To once more highlight another important aspect, it has been commonplace that nominal concord or agreement of sorts is a precondition for LBE essentially since Ross (1967). However, morphologically manifested nominal concord obtains even in MR in terms of number and gender, clearly showing that in and of itself, nominal concord is not enough for the licensing of LBE:

- (55) a. l-a plante vert-e
 the-F.SG plant.F.SG green-F.SG
 ‘the green plant’
- b. l-e bâtiment vert
 the-M.SG building.M.SG green.M.SG
 ‘the green building’ (*French*; Norris 2017)

What seems to be required is rich noun inflection – the crucial property that distinguished Latin from MR. The current approach in terms of n_{str} and n_{wk} accommodates these facts naturally in that the former underlies a rich and varied nominal case paradigm while the latter underlies an absent case paradigm.

In his discussion on discontinuous noun phrases, Pinkster (2021, 1098) recently remarks that discontinuity is not contingent on agreement as evidenced by examples like (56) where a postnominal genitive attribute undergoes fronting, which does not agree with the head noun:

- (56) Quis enim est qui putet ... **nostrorum hominum** ad eorum doc-

tissimas voces **aures** clausas fuisse?

‘For who can imagine that the ears of our countrymen were closed to the echo of their wisdom?’ (Cic. Tusc. 4.3.)

A remark is in order, and a question. The example (56) is not counterevidence to the claim made in this paper that LBE is licensed by rich agreement generally, and rich noun inflection in particular. As Pinkster (2021, 1098) points out, the NP *nostrorum hominum* shows up postnominally in the contiguous variant and thus does not fall under the rubric LBE. Secondly, why Latin allowed postnominal genitive attributes to be extracted from NPs is an interesting question. Languages appear to vary with respect to this possibility. Thus Serbo-Croatian (57) and German (58) disallow it, while Polish fully (59) and Czech marginally allows it (60):

- (57) a. ?*[Ovog studenta]_i sam pronašla [knjigu _{t_i}]
 this.GEN student.GEN am found book.ACC
 ‘Of this student I found the book.’
 b. *Koga_i si pronašla [knjigu _{t_i}]
 who.GEN are found book.ACC
 ‘Of whom did you find the book?’ (Serbo-Croatian, Zlatić
 1997 in Bošković 2013b)
- (58) a. *[Dieses Studenten]_i habe ich [ein Buch _{t_i}] gefunden.
 this.GEN student.GEN have I a book found
 ‘I found a book of this student.’
 b. *Wessen_i hast du [ein Buch _{t_i}] gefunden?
 whose.GEN have you a book found
 ‘Of whom did you find a book?’ (German)

- (59) a. On kupił [[tomik [poezji angielskiej]] [w skórzanej
he purchased volume poetry.GEN English.GEN in leather
okładce]].
jacket
‘He purchased a volume of English poetry with a leather jacket.’
- b. [Jakiej poezji]_i on kupił [[tomik *t_i*] [w skórzanej
what.GEN poetry.GEN he purchased volume in leather
okładce]]?
cover
‘What poetry did he purchase a volume of with a leather jacket?’
- (*Polish*, Szczegielniak 2021)
- (60) ?(?) [Kojeg doktora]_i si vidio [prijatelja *t_i*]
which doctor.GEN are seen friend.ACC
‘Which doctor did you see a friend of?’ (*Czech*, Starke 2001, 39 in
Bošković 2018)

The emerging picture with respect to the extraction of complements of nouns is inconsistent: The article-less languages Serbo-Croatian, Czech, Polish and Latin exhibit rich noun inflection but differ with respect to extracting postnominal genitive NPs. The article-language German has poor nominal inflection and appears to pattern with Serbo-Croatian.

A reviewer points out an important issue:

“genitive can be an inherent Case (it is the structural case assigned by nouns in Slavic but it is also clear that there are instances of inherent genitive in Slavic). If [(56)] is not fully productive maybe it is an instance of inherent case?”

An investigation of this question must be left for future research both with respect to the Latin case as well as the Polish one. For more cases from

Greek, cf. Bošković (2015) with a different account which blocks extraction of genitive complements quite generally.

Example (60) involves extraction of the genitive NP from a structurally case marked NP and is chosen to contrast with an inherently case marked NP. The judgment is thus possibly to be taken within this context and arguably needs to be idealized. Concluding that the richness of nominal inflection does not figure in the licensing of this possibility is thus premature, especially in light of the Polish facts from (59). Richer and more careful data elicitation and argumentation is required. Why Polish does, and Latin did, permit extraction of genitive NPs remains to be understood.

4.3 A discussion

Let us take stock. In this paper, I suggest a uniform analysis of the distribution of DET-categories in Latin and MR couched in the POP/+ framework (Chomsky 2013, 2015). The analysis has the following characteristics: Merge applies freely and untriggered, and there is no projection. Instead, an interface condition on labeling (the LR) and 3rd factor principle of efficient computation interact to deliver the label for a given set formed by Merge. Following Blümel (2024); I have adopted the Nominal Strength Parameter which states that *n* is parametrized as strong (Latin) and weak (MR). The diachrony is related to the loss of morphological nominal inflection, indicating the underlying replacement of strong *n* by weak *n*. The approach has many implications of which I can address only one here, diverting a fuller discussion to a different occasion (cf. Blümel 2025); I will confine myself to diachronic aspects in this discussion.

At least as far as the diachrony of the syntax of the nominal structure in Romance goes, the current approach casts doubt on economy principles that attribute historical clines on preferences of head over phrase status as forcefully suggested in much recent work by van Gelderen (2004, 2018, 2019) and Dadan (2019). van Gelderen (2004, 18) suggests the Head Preference Principle given in (61):

(61) **Head Preference Principle**

Be a head, rather than a phrase.

A commonality of these works is that *Minimal Search* exerts a pressure on structures of the form XP-YP such that over time, they gave way to Head-XP structures. In the nominal domain, this is played out by demonstratives turning into articles over time. By assumption, the former are phrasal and the latter are heads. Thus, there was a change from demonstratives in Old English (a language lacking articles) towards stages in which it assumed an article system over time, illustrated in (62).

- (62) a. Brittene igland is ehta hund mila lang. & twa hund brad. & her sind on **þis** iglande fif geeode. Englisc. & Brittisc. & Wilsce. & Scyttisc. & Pyhtisc. & Boc Leden.
 ‘The island Britain is 800 miles long and 200 miles broad. And there are in the island five nations; English, Welsh, Scottish, Pictish, and Latin.’ Old English-style in the Peterborough Chronicle
- b. is gære for **þe** king Stephne ofer sæ to Normandi; and ther wes

underfangen forþi at hi uuenden at he sculde ben alsuic alse
the eom wes,
 ‘This year, (the) King Stephen crossed the sea to go to Nor-
 mandy and was received there because they thought he was like
 the uncle.’ 1137

van Gelderen (2019) adumbrates the diachronic development shown in (63) for this change; the DP on the left side of the arrow represents the demonstrative.

$$(63) \quad [? [_{DP} D] [_{NP} N \dots]] \rightarrow [_{DP} D [_{NP} N \dots]]$$

The former is an XP-YP structure while the latter is a Head-XP structure. The former allegedly sits uneasily with *Minimal Search* while the latter obeys the condition.

The current work seeks to capture a similar historical development, but the analysis is replaced by a structure in which the target is assimilated to the source: Both Latin and MR feature DET-categories in “SPEC” of n . Crucially, in the current approach the DP-hypothesis is abandoned and articles are phrasal. If the current analysis can be sustained, the claim that *Minimal Search* is the driving force behind diachronic change cannot be quite right, because DET-categories are invariably phrasal and there is no “grinding off” of specifiers to yield heads. While the forms and surface manifestations of DET-categories are subject to change, the key underlying change concerns n with ramifications in the DET-system: In the n_{wk} -languages of MR DET became obligatory to ensure a label.

Why did articles evolve and why did Latin lack them? The NSP predicts an obligatory XP in the sister position of $[_\alpha \text{ } n_{wk} \text{ R}]$ and the LBC. The NSP is consistent with the generalization given in (64) (cf. Uriagereka 1988; Corver 1989; Bošković 2005a) – however, it does not strictly predict it:

- (64) Only languages without articles may allow LBE. (slightly modified from Bošković 2005a)

A possible route at an explanation of (64) is this: Articles diachronically emerge only in languages with n_{wk} ; i.e., in complex nominals with an EPP-effect, perhaps analogous to expletives in the clause. For lexically designated expletives, there is evidence that they only indirectly relate to EPP-properties of different kinds and can evolve with a delay.¹¹ If similar diachronic paths can be established within complex nominals, the correlation between the availability of articles and the LBC could be captured: the nominal EPP is a precondition for the development of articles.

There is manifold evidence that MR features what is sometimes called expletive articles (Longobardi 1994). Thus, the definite articles in (65) precede common count nouns in the inalienable construction.¹² The construction receives a bound reading. In (66) the definite article introduces a proper name. Finally, in (65-b) the definite articles precedes an inalienable noun:

¹¹E.g., cf. Fuß 2018 on the historical development the prefield expletive *es* ‘it’ to satisfy the verb second constraint in Middle High German.

¹²Likewise, Ticio (2005) observes that Spanish does not show a definiteness effect when a theme is extracted out of a syntactically complex nominal headed by a definite article – cases which are sometimes treated in terms of expletive definite article.

More generally, it would be interesting to see how those cases developed diachronically in the Latin-to-MR change as a reviewer points out to me.

- (65) a. Les enfants ont levé **la** main.
the children have raised the hand
The children raised their hand.’ (*French*, Vergnaud and
Zubizarreta 1992, 596)
- b. Pedro se lavó **la** cara esta mañana.
Pedro CL washed the face this morning
‘Pedro washed his face this morning.’ (*Spanish*, Espinal and
Cyrino 2017, 2)
- (66) **Il** Gianni mi ha telefonato.
the Gianni me has called
‘Gianni called me up.’ (*Italian*, Longobardi 1994, 622)

Still another option is suggested by a kind of morphological trade-off the diachrony of nouns gives rise to. A weaker version of such a hypothesis was suggested to me by a reviewer who states: “why, when present, [do] articles typologically tend to be those that carry case agreement, as in German?” (67) illustrates this:

- (67) German ‘the table’
- a. **der** Tisch
D-NOM table
- b. **den** Tisch
D-ACC table
- c. **dem** Tisch
D-DAT table
- d. **des** Tisches
D-GEN table

(*German*)

The current paper entertains the hypothesis that in the transition from Latin to MR n_{str} is replaced by n_{wk} such that the presence of a DET-category in the sister position of weak nP is enforced. Simplifying, this goes hand in hand with a loss of case as well as gender marking on nouns. Now as far as articles are concerned, these could develop as the formal elements that compensate for this loss in that they bear case or gender marking. In effect, DET takes over the strength of n ; schematically (I here remain neutral with respect to the question of the labeling-properties of DET in the initial stage):

$$(68) \quad (\text{DET}) \ n_{str} \rightarrow *(\text{DET}_{str}) \ n_{wk}$$

Plausibly, this shift of strength from n to DET takes place in languages having nominal φ -features at all and retain them on n . That is, no such shift is expected in languages which lack nominal φ -features. Given that the morphological expression of φ diachronically disappears on n ; DET takes over this function as a default. Given that most –if not all– DET-categories have a semantic or grammatical meaning other than expressing φ , definite articles evolve as categories on their own right to morphologically manifest φ . Furthermore, the German facts suggest that definite articles are decomposable into $\{\varphi_{str}, [\text{DEF}]\}$ in languages where definiteness is grammatically encoded.¹³

¹³What about languages like English, which lack strong morphological manifestation on determiners? Evidence for the strength of DET could be manifested indirectly in such languages. For instance, it is conceivable that fusional and suppletive morphology in the pronominal system of the language indicates the strong character of DET in such languages, assuming that pronouns belong to this category (cf. Postal 1966). Such systems would then contrast with agglutinating strategies found in languages like Japanese and Mandarin Chinese.

I can only give this speculation here. Notice that I interpret the obligatory presence of DET in n_{wk} -languages as an EPP-effect in the nominal domain. Articles can thus be considered a special strategy to fulfill the DET-requirement which n_{wk} gives rise to. As there are analogues in the clausal domain, such developments could be investigated in tandem.

5 Conclusion

The analysis I propose in this paper sheds new light on “Latin’s recourse to synthetic strategies, in contrast to those of a predominantly analytic nature in Romance” (Ledgeway 2012, 424): The historical development from a language with rich noun inflection (rich case paradigms and declension classes), i.e., Latin, to a set of languages with poor or no noun inflection, i.e., Modern Romance, is reflected in the Nominal Strength Parameter. While Merge applies freely, its output representations have to abide by the interface condition that every syntactic object be labeled by *Minimal Search*. This is met in the n_{str} -languages Latin without ado with concomitant optionality of determiners and the option of Left Branch Extraction. The Labeling Requirement can be met in n_{wk} -languages (modern Romance) only if $\{n; R\}$ is accompanied by xAP to yield $\{xAP, \{n; R\}\}$; deriving the obligatoriness of determiners and the effect of the Left Branch Condition. Loss of case morphology was thus a reflection of the parametric shift from n_{str} to n_{wk} ; the underlying cause of rather dramatic syntactic changes in the nominal domain.

References

- Agbayani, B. and C. Golston (2016). Phonological constituents and their movement in latin. Phonology 33(1), 1–42.
- Bauer, B. L. M. (2009). Word order. In P. Baldi and P. Cuzzolin (Eds.), New perspectives on historical Latin syntax (Trends in Linguistics. Studies and Monographs [TiLSM] 180), Volume 1, pp. 241–316. Berlin: Mouton de Gruyter.
- Blümel, A. (2024). Labeling Theory and the Nominal Phrase. In J. Lu, E. Petersen, A. Zaitsev, and B. Harizanov (Eds.), Proceedings of WCCFL 40.
- Blümel, A. (2025). Strength and Weakness in the Nominal Domain – Evidence from the Distribution of DET and N. Ms. HU Berlin and University of Göttingen.
- Blümel, A. and A. Holler (2022). DP, NP, or neither? Contours of an unresolved debate. Glossa – a journal of general linguistics 7(1).
- Borer, H. (2005). In Name Only. Structuring Sense, Volume I. Oxford University Press.
- Bošković, Ž. (2005a). Left branch extraction, structure of NP, and scrambling. In J. Sabel and M. Saito (Eds.), The free word order phenomenon: Its syntactic sources and diversity, pp. 13–73. Mouton de Gruyter, Berlin.
- Bošković, Ž. (2005b). On the locality of left branch extraction and the structure of NP. Studia Linguistica 59(1–4).

- Bošković, Ž. (2008). What will you have, DP or NP? In Proceedings of NELS 37.
- Bošković, Ž. (2012). On NPs and Clauses. In G. Grewendorf and T. E. Zimmermann (Eds.), Discourse and grammar: From sentence types to lexical categories, pp. 179–242. De Gruyter.
- Bošković, Ž. (2013a). Adjectival escapades. In In Proceedings of FASL 21, pp. 1–25.
- Bošković, Ž. (2013b). Phases beyond clauses. In L. Schürcks, A. Giannakidou, U. Etxeberria, and P. Kosta (Eds.), Nominal Constructions in Slavic and Beyond, pp. 75–128. Mouton de Gruyter, Berlin.
- Bošković, Ž. (2015). From the complex np constraint to everything: On deep extractions across categories. The Linguistic Review (GLOW issue) 32, 603–669.
- Bošković, Ž. (2016). On second position clitics crosslinguistically. In F. Marušič and R. Žaucer (Eds.), Formal studies in Slovenian syntax: In honor of Janez Orešnik, pp. 23–54. Amsterdam: John Benjamins.
- Bošković, Ž. (2018). On Extraction out of Inherently Case-Marked Elements. In Proceedings of FASL 25, pp. 57–78.
- Chomsky, N. (1973). Conditions on Transformations. In S. R. Anderson and P. Kiparsky (Eds.), A Festschrift for Morris Halle, Number 232-286. New York: Holt, Rinehart and Winston.
- Chomsky, N. (1986). Knowledge of language. New York: Praeger.

- Chomsky, N. (1995). The Minimalist Program. Cambridge, Massachusetts: MIT Press.
- Chomsky, N. (2004). Beyond Explanatory Adequacy. In A. Belletti (Ed.), Structures and Beyond, pp. 104–131. Oxford: Oxford University Press.
- Chomsky, N. (2007). Approaching UG from Below. In U. Sauerland and H.-M. Gärtner (Eds.), Interfaces + Recursion = Language?: Chomsky’s Minimalism and the View from Syntax-Semantics, pp. 1–29. Berlin, New York: Mouton de Gruyter.
- Chomsky, N. (2008). On Phases. In R. Freidin, C. P. Otero, and M. L. Zubizarreta (Eds.), Foundational Issues in Linguistics, pp. 133–166. Cambridge, Massachusetts: MIT Press.
- Chomsky, N. (2013). Problems of Projection. Lingua 130, 33–49.
- Chomsky, N. (2015). Problems of Projection: Extensions. In E. D. Domenico, C. Hamann, and S. Matteini (Eds.), Structures, strategies and beyond – studies in honour of Adriana Belletti, pp. 3–16. John Benjamins.
- Corver, N. (1989). Left branch extractions and DP. In H. Bennis and A. van Kemenade (Eds.), Linguistics in the Netherlands, pp. 31–40. Foris: Dordrecht.
- Corver, N. (1990). The Syntax of Left Branch Extractions. Ph. D. thesis, Tilburg University.
- Corver, N. (2010). Subextraction. ms. University of Utrecht.

- Dadan, M. (2019). Head Labeling Preference and Language Change. Ph. D. thesis, University of Connecticut.
- D’Alessandro, R. (2015). Null subject. In A. Fábregas, J. Mateu, and M. Putnam (Eds.), Contemporary linguistic parameters, pp. 201–226. London: Bloomsbury Press.
- Danckaert, L. (2017). The Development of Latin Clause Structure: A Study of the Extended Verb Phrase. Oxford: Oxford University Press.
- Delfitto, D. and J. Schroten (1991). Bare plurals and the number affix in DP. Probus 3(2), 155–185.
- Devine, A. and L. Stephens (2000). Discontinuous syntax. Hyperbaton in Greek. Oxford: Oxford University Press.
- Devine, A. M. and L. D. Stephens (2006). Latin Word Order. Structured Meaning and Information. New York: Oxford University Press.
- Epstein, S., H. Kitahara, and D. Seely (2014). Labeling by Minimal Search: Implications for successive cyclic A-movement and the conception of the postulate ‘phase’. Linguistic Inquiry 45(3), 463–481.
- Epstein, S. D., H. T. Kitahara, and D. S. Seely (2020). Unifying Labeling under Minimal Search in “Single-” and “Multiple-Specifier” Configurations. Koyote Papers: Working Papers in Linguistics, Linguistic Theory at the University of Arizona 22, 1–11.
- Espinal, M. and S. Cyrino (2017). The definite article in romance expletives

and long weak definites. Glossa: a journal of general linguistics 2(1)(23), 1–26.

Espinal, M. T. (2010). Bare nominals in Catalan and Spanish. their structure and meaning. Lingua (120), 984–1009.

Feldcamp, Z. S. (2021). The Split Noun Phrase in Classical Latin. Ph. D. thesis, The University of Arizona.

Francom, J. (2012). Wh-movement: Interrogatives, exclamatives, and relatives. In J. I. Hualde, A. Olarrea, and E. O’Rourke (Eds.), Handbook of Hispanic Linguistics, First Edition. Blackwell Publishing.

Franks, S. (1995). 1995: Parameters of Slavic Morphosyntax. New York/Oxford: Oxford University Press.

Franks, S. (2007). Deriving discontinuity. In Proceedings of FDSL 6.5, pp. 102–120.

Fuß, E. (2018). Vorfeld-expletives in the history of German. Talk at the Workshop “From Sentence Grammar to Discourse Grammar and from Discourse Grammar to Sentence Grammar“, University of Göttingen.

Goto, N. (2017). Eliminating the Strong/Weak Parameter on T. In M. Y. Erlewine (Ed.), Proceedings of GLOW in Asia XI, MIT Working Papers in Linguistics, Volume 2 of MIT Working Papers in Linguistics.

Grosu, A. (1974). On the nature of the left branch condition. Linguistic Inquiry 5(2), 308–319.

- Hayashi, N. (2020). Labeling without Weak Heads. Syntax 23(3), 275–294.
- Horn, G. (1978). Toward a More Adequate Definition of the Notion of Transformation. Edmonton: Linguistic Research.
- Horn, G. (1983). Lexical-Functional Grammar. Berlin: Mouton de Gruyter.
- Jakobson, R. (1984). Contribution to the general theory of case: General meanings of the russian cases. In Russian and Slavic Grammar: Studies 1931-1981, pp. 59–103. Berlin/New York/Amsterdam: Mouton.
- Kruger, W. (2017). Free Merge, Delayed Agree, and Phase-Cancellation: An Account of (Anti-)That-Trace Effects. In A. Lamont and K. Tetzloff (Eds.), Proceedings of the Forty-Seventh Meeting of the North East Linguistics Society (NELS47), Volume 2, pp. 191–200. UMass: Amherst.
- Laury, R. (1997). Demonstratives in interaction. The emergence of a definite article in Finnish. Amsterdam: Benjamins.
- Ledgeway, A. (2012). From Latin to Romance: The Rise of Configurationality, Functional Categories and Head-marking. In J. Baral, M. Cenamo, and E. van Gelderen (Eds.), Variation and Change in Argument Realisation, Volume 110, pp. 422–442. Oxford: Blackwell. Special Issue of the Transactions of the Philological Society.
- Leu, T. (2008). The Internal Syntax of Determiners. Ph. D. thesis, New York University.
- Leu, T. (2014). The Architecture of Determiners. Oxford University Press.

- Longobardi, G. (1994). Reference and proper names. Linguistic Inquiry 25, 609–66.
- Loporcaro, M. (2017). Gender from Latin to Romance. Oxford: Oxford University Press.
- Marantz, A. (1997). No Escape from Syntax: Don’t Try Morphological Analysis in the Privacy of Your Own Lexicon. In Working Papers in Linguistics, Philadelphia, Proceedings of the 21st Penn Linguistics Colloquium, pp. 201–225.
- McInnerney, A. (2024). The Position of Wh-Subjects in Labeling Theory. Linguistic Inquiry 55(3), 579–594.
- Norris, M. (2014). A theory of nominal concord. Ph. D. thesis, University of California, Santa Cruz.
- Norris, M. (2017). Description and analyses of nominal concord (Pt I). Language and Linguistic Compass 11(11).
- Obata, M. (2018). Eliminating C-deletion in the syntax: structure-building by merge. Koganei Journal of the Humanities, 21–34.
- Obata, M. and S. D. Epstein (2011). Feature-splitting internal merge: Improper movement, intervention, and the a/a distinction. Syntax 14(2).
- Oda, H. (2022). The NP/DP-language Distinction as a Scale and Parameters in Minimalism. Ph. D. thesis, University of Connecticut.

- Oishi, M. (2015). The Hunt for a Label. In H. Egashira, H. Kitahara, K. Nakazawa, T. Nomura, M. Oishi, A. Saizen, and M. Suzuki (Eds.), Untiring Pursuit of Better Alternatives, pp. 322–334. Kaitakusha, Tokyo.
- Oseki, Y. (2014). Eliminating Pair-Merge. Handout of the 32nd West Coast Conference on Formal Linguistics.
- Ott, D. (2011). Local instability: The syntax of split topics. Ph. D. thesis, Cambridge, MA: Harvard University.
- Pankau, A. (2019). Left branch extraction in Lower Sorbian – A problem for the DP/NP analysis. Talk at the FASL 28, Stony Brook University, 03.-05.05.2019.
- Pinkster, H. (1990). The development of cases and adpositions in Latin. In H. Pinkster and I. Genée (Eds.), Unity in diversity. Papers presented to Simon C. Dik on his 50th birthday, pp. 195–209. Dordrecht: Foris.
- Pinkster, H. (2021). The Oxford Latin Syntax: The Complex Sentence and Discourse. Oxford University Press.
- Postal, P. (1966). On so-called “pronouns” in English. In F. Dinneen (Ed.), Report of the Seventeenth Annual Round Table Meeting on Linguistics and Language Studies, pp. 177–206. Washington, D.C.: Georgetown University Press.
- Powell, J. (2010). Hyperbaton and register in Cicero. In E. Dickey and A. Chahoud (Eds.), Colloquial and Literary Latin, pp. 163–185. Cambridge: Cambridge University Press.

- Reeve, M. (2019). An agreement-based analysis of extraction from nominals. Natural Language and Linguistic Theory 37(1), 263–314.
- Rizzi, L. (1982). Issues in Italian syntax. Dordrecht: Foris.
- Rizzi, L. and U. Shlonsky (2007). Strategies of subject extraction. In H.-M. Gärtner and U. Sauerland (Eds.), Interfaces + Recursion = Language, pp. 115–60. Berlin: Mouton de Gruyter.
- Ross, J. (1967). Constraints on Variables in Syntax. Ph. D. thesis, MIT.
- Ross, J. (1986). Infinite syntax! Norwood, NJ: ALEX.
- Sag, I., T. Wasow, and E. Bender (2003). Syntactic theory: a formal introduction. Stanford, CA: CSLI Publications.
- Schwegler, A. (1990). Analyticity and syntheticity. A diachronic perspective with special reference to Romance languages. Berlin: Mouton de Gruyter.
- Spevak, O. (2010). Constituent Order in Classical Latin Prose. Number 117 in Studies in Language Companion Series (SLCS). Amsterdam/Philadelphia: John Benjamins Publishing Company.
- Starke, M. (2001). Move dissolves into Merge: A theory of locality. Ph. D. thesis, University of Geneva.
- Szabolcsi, A. (1983). The possessor that ran away from home. The Linguistic Review 3(1), 89–102.
- Szczegielniak, A. (2021). DP’s in Polish. Presentation at the SLS Meeting Urbana-Champaign 2021.

- Ticio, M. E. (2005). Locality and anti-locality in spanish dps. Syntax 8(3), 229–286.
- Uriagereka, J. (1988). On government. Ph. D. thesis, University of Connecticut.
- van Gelderen, E. (2004). Specifiers, heads, grammaticalization, and economy. Journal of Comparative Germanic Linguistics 7, 59–98.
- van Gelderen, E. (2018). Problems of projection: The role of language change in labeling paradoxes. Studia Linguistica 72(1), 113–127.
- van Gelderen, E. (2019). Cyclical change and problems of projection. In A. Breitbarth, M. Bouzouita, L. Danckaert, and E. Witzelhausen (Eds.), Cycles in Language Change, pp. 13–32. Oxford University Press.
- Vergnaud, J. R. and M. L. Zubizarreta (1992). The definite determiner and the inalienable constructions in French and in English. Linguistic Inquiry 23(4), 595–652.
- Viti, C. (2010). Observations on genitive word order in latin. In O. Spevak (Ed.), Le syntagme nominal en latin. Nouvelles contributions, pp. 77–96. Paris, Harmattan.
- Wigtil, D. N. (1993). Latin Definiteness and English Articles. The Classical world 86(6), 467–481.
- Yanagisawa, K. (2021). Subject extraction and the two EPP requirements. Explorations in English Linguistics (35), 89–110.

Zlatić, L. (1997). The Structure of the Serbian Noun Phrase. Ph. D. thesis,
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